

TrendWatch AI — Social Media Intelligence Report

Use Case 2: Brand monitoring on Reddit · Generated 2026-06-09 03:36 UTC

Query: meta r/technology Limit: 45 Time: devvit_import

Data: Live Reddit (imported from Devvit app)

Report contents (assignment sections)

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1. Executive briefing

Source: rule_based_fallback · RAG + CoT LLM or rule-based fallback

Executive summary

TrendWatch AI analysed 45 Reddit posts. Negative sentiment is 37.8%. Crisis score is 36/100 (Medium).

Main concerns

- Risk keywords appeared 4 times across posts.
- 17 high-engagement posts carry negative sentiment.

Crisis explanation

Risk keywords appeared 4 times across posts. 17 high-engagement posts carry negative sentiment.

Recommended actions

- Monitor high-impact negative threads daily.
- Prepare a factual FAQ for top complaint topics.
- Escalate to customer support if refund/delay keywords spike.

Communication strategy

Address emerging topics: Meta / After / Says, Meta / Instagram / Accounts, Meta / Smart / Glasses. Use empathetic, transparent tone; avoid dismissive responses.

Full LLM report (markdown)

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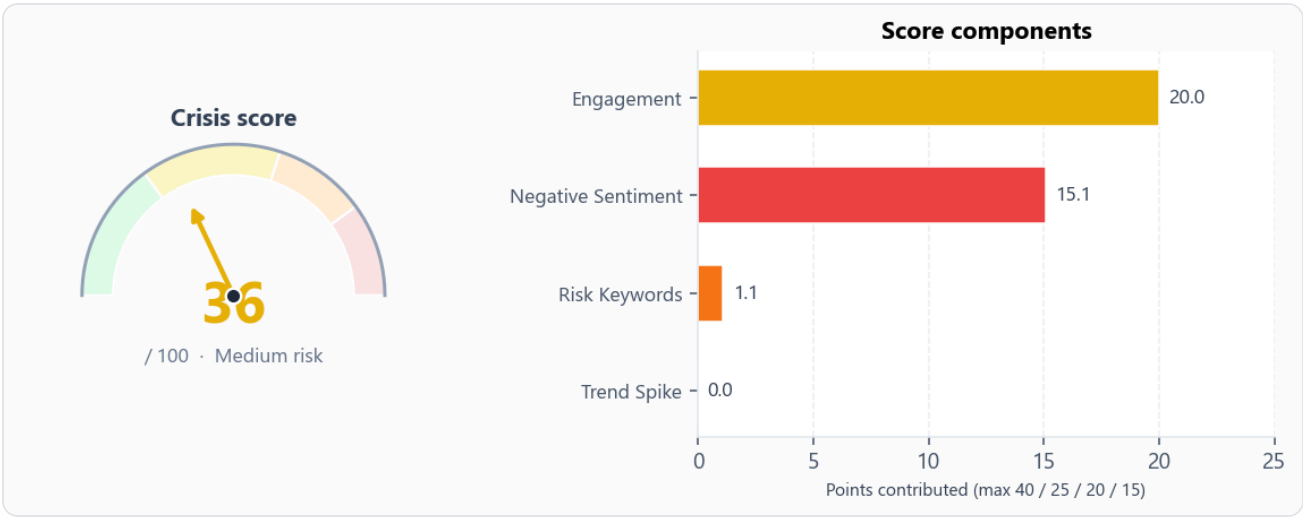
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Communication Strategy

Address emerging topics: Meta / After / Says, Meta / Instagram / Accounts, Meta / Smart / Glasses. Use empathetic, transparent tone; avoid dismissive responses.

Ethics notice: Analysis uses public Reddit text. Sentiment and crisis scores are algorithmic estimates, not legal or PR advice. Models may reflect language bias; human review is required before brand actions.

2. Crisis intelligence layer



36/100 — Medium risk

Weighted fusion of negative sentiment, risk keywords, high-engagement negative posts, and trend spikes.

Score components

Component	Points
negative_sentiment	15.1
risk_keywords	1.1
engagement	20.0
trend_spike	0.0

Reasons

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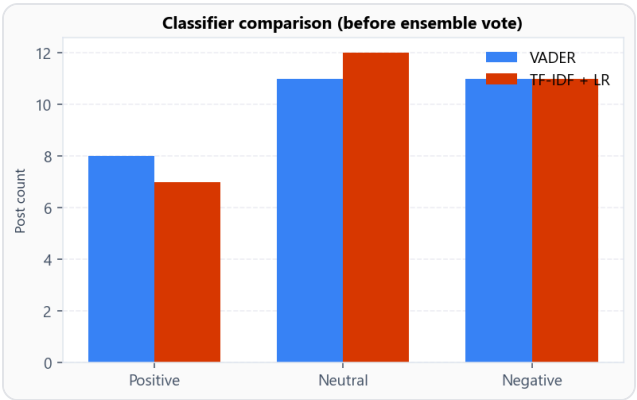
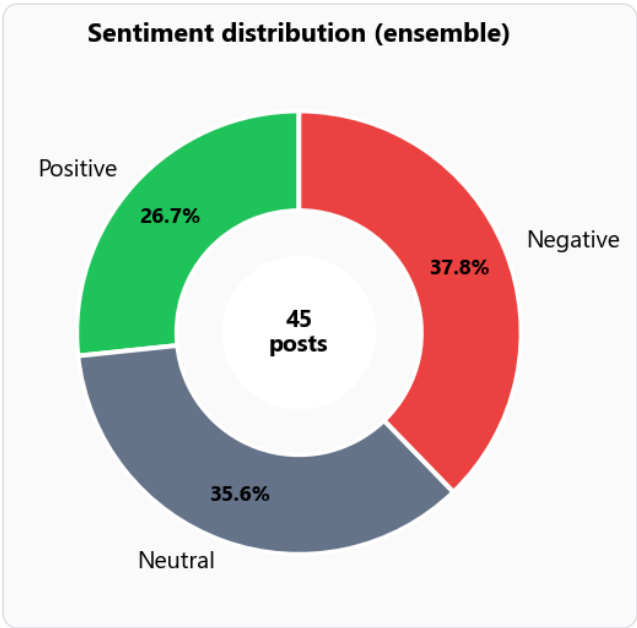
3. Overview & sentiment analysis

45
POSTS ANALYSED

26.7%
POSITIVE

35.6%
NEUTRAL

37.8%
NEGATIVE



Independent classifier outputs before ensemble fusion.

Final labels after VADER + LR ensemble majority vote.

Sentiment distribution (VADER + LR ensemble)

Positive26.7%

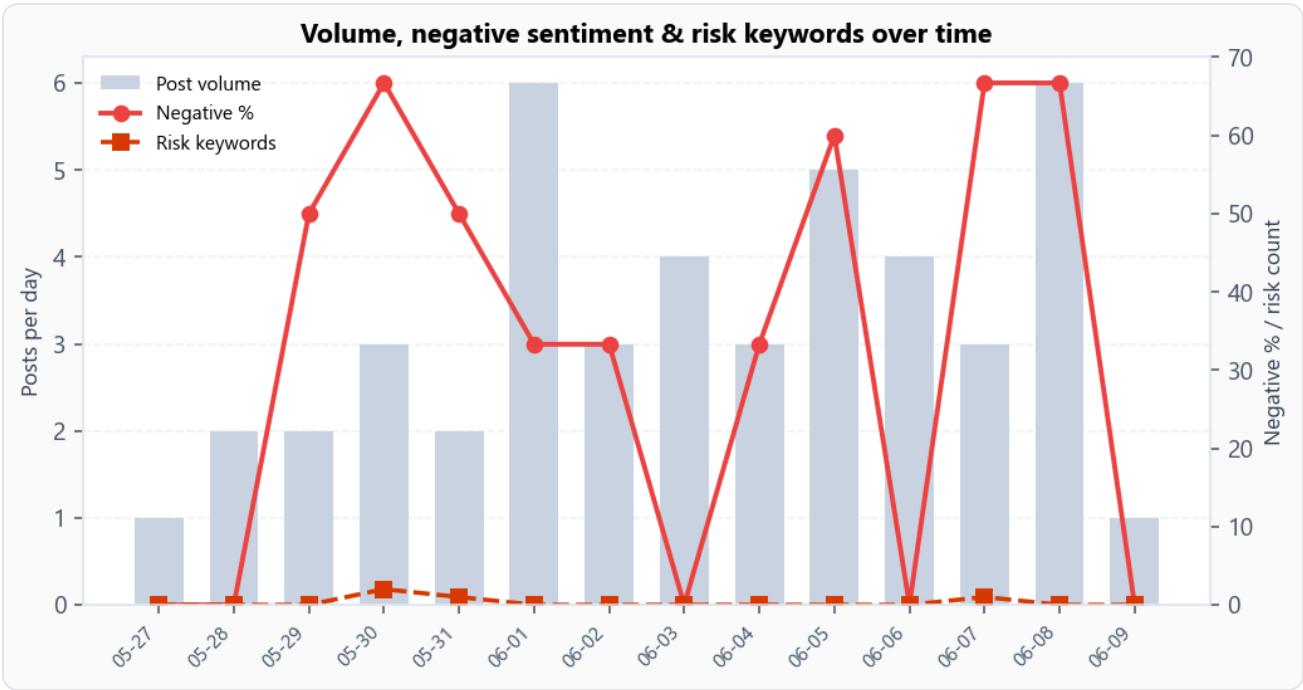
Neutral35.6%

Negative37.8%

Label	Count
positive	12
neutral	16
negative	17

Counts: positive=12, neutral=16, negative=17

4. Trend detection



Daily post volume (bars), negative sentiment % (line), and risk keyword counts (dashed).

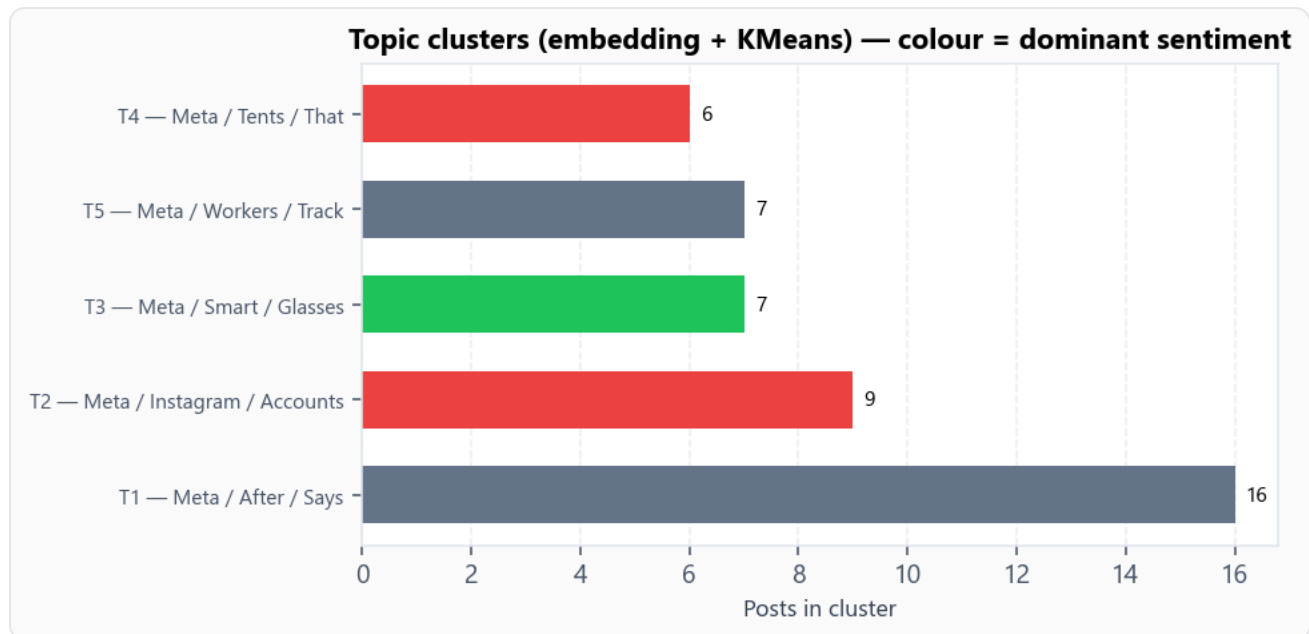
Spike detected: No — No significant negative sentiment spike detected.

Average negative ratio: 32.9%

Date	Posts	Negative	Neg %	Risk keywords
2026-05-27	1	0	0.0%	0
2026-05-28	2	0	0.0%	0
2026-05-29	2	1	50.0%	0
2026-05-30	3	2	66.7%	2
2026-05-31	2	1	50.0%	1
2026-06-01	6	2	33.3%	0
2026-06-02	3	1	33.3%	0
2026-06-03	4	0	0.0%	0
2026-06-04	3	1	33.3%	0
2026-06-05	5	3	60.0%	0

Date	Posts	Negative	Neg %	Risk keywords
2026-06-06	4	0	0.0%	0
2026-06-07	3	2	66.7%	1
2026-06-08	6	4	66.7%	0
2026-06-09	1	0	0.0%	0

5. Topic clusters (embeddings + KMeans)



Topic 0: Meta / After / Says

Size: 16 · Avg sentiment: **neutral**

Keywords: meta, after, says, canada, metadata, weighs

Representative: Signal, DuckDuckGo, and NordVPN threaten to exit Canada if metadata surveillance law passes

Topic 1: Meta / Instagram / Accounts

Size: 9 · Avg sentiment: **negative**

Keywords: meta, instagram, accounts, hackers, access, into

Representative: Hackers Simply Asked Meta AI to Give Them Access to High-Profile Instagram Accounts. It Worked

Topic 2: Meta / Smart / Glasses

Size: 7 · Avg sentiment: **positive**

Keywords: meta, smart, glasses, recognition, face, code

Representative: Meta Furious Over Bombshell Smart Glasses Revelation

Topic 4: Meta / Workers / Track

Size: 7 · Avg sentiment: **neutral**

Keywords: meta, workers, track, clicks, train, employee

Representative: Meta's AI training effort is capturing employee emails and browsing history, not just mouse clicks

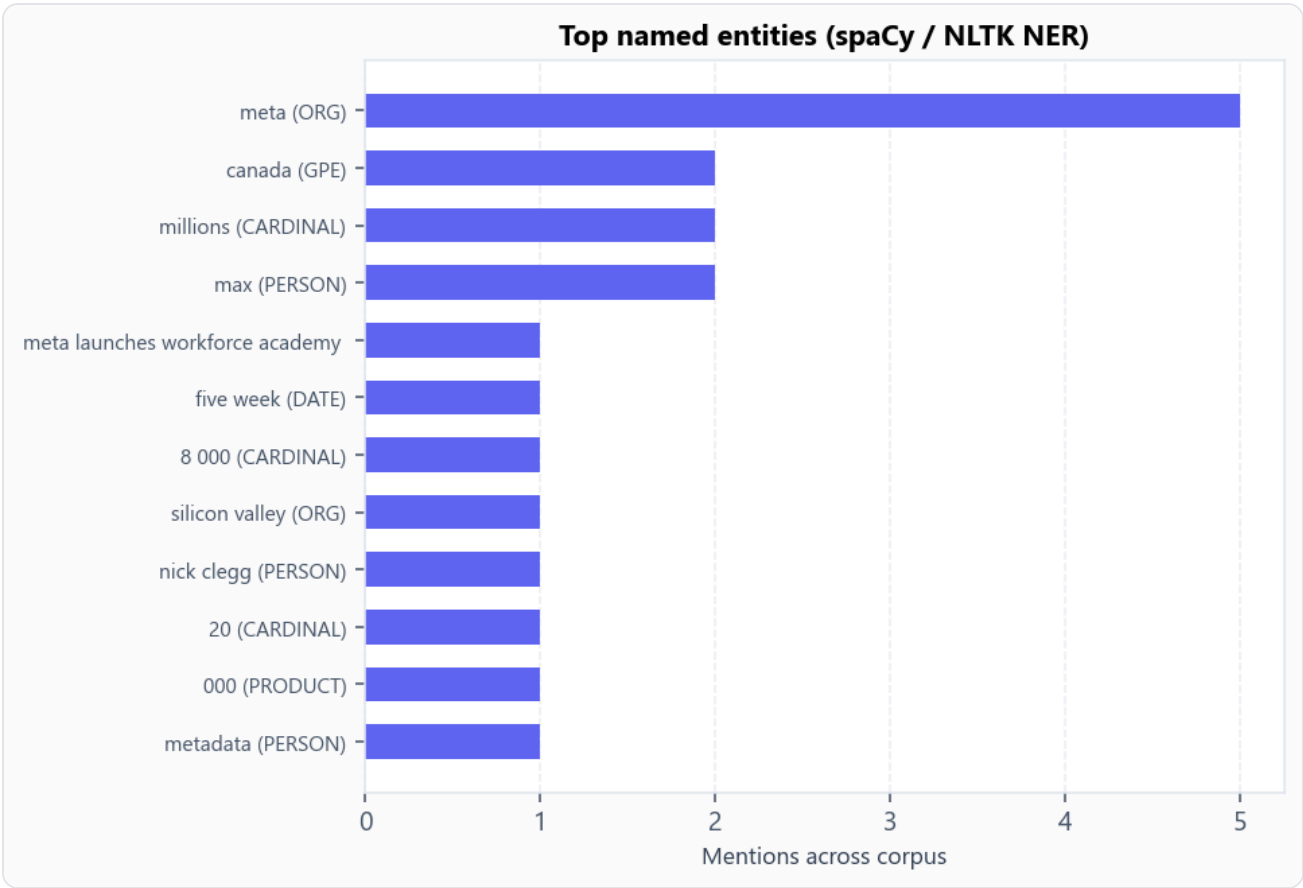
Topic 3: Meta / Tents / That

Size: 6 · Avg sentiment: **negative**

Keywords: meta, tents, that, data, centers, just

Representative: Meta So Desperate for Compute That It's Building "Data Centers" That Are Just Tents Filled With AI Chips

6. Named entities (NER)



Entity	Mentions
meta (ORG)	5
canada (GPE)	2
millions (CARDINAL)	2
max (PERSON)	2
meta launches workforce academy (ORG)	1
five week (DATE)	1
8 000 (CARDINAL)	1
silicon valley (ORG)	1
nick clegg (PERSON)	1
20 (CARDINAL)	1

Entity	Mentions
000 (PRODUCT)	1
metadata (PERSON)	1
two thirds (CARDINAL)	1
brussels (GPE)	1
meta s smart (ORG)	1

7. Risk keywords & high-impact posts

Risk keyword frequency

Keyword	Count
privacy	2
fraud	1
lawsuit	1

Similar post pairs (cosine similarity)

Post A	Post B	Similarity
Meta AI chatbot enabled hackers to access others' Insta	t3_1tu2pwj	0.936
Meta Silently Added Face-Recognition Code for Its Smart	Meta Silently Added Face-Recognition Code for Its Smart	0.934
Meta Says 20,000 Instagram Accounts Hacked via AI Tool	t3_1ttnq7	0.823
t3_1tu2pwj	t3_1ttnq7	0.815
Meta AI chatbot enabled hackers to access others' Insta	t3_1ttnq7	0.806

High-impact posts

Title	Subreddit	Score	Comments	Sentiment	Risk terms	Impact
Meta Furious Over Bombshell Smart Glasses	technology	10537	762	negative	—	15.9

Title	Subreddit	Score	Comments	Sentiment	Risk terms	Impact
Revelation						
Signal, DuckDuckGo, and NordVPN threaten to exit Canada if metadata surveillance...	technology	16655	429	negative	—	15.78
Women in Brussels 'filmed without their knowledge' by men wearing Meta smart gla...	technology	8584	733	positive	—	15.66
Meta Deletes Face-Recognition System From Its Smart Glasses App After WIRED Repo...	technology	8730	277	positive	—	14.7
Meta So Desperate for Compute That It's Building "Data Centers" That Are Just Te...	technology	5840	367	negative	—	14.58

8. Post-level analysis (sample)

Showing up to 30 posts with VADER label, LR label, and extracted risk terms.

Title	Author	Ensemble	VADER	LR	Risk terms
Meta launches 'Workforce Academy' to train workers to build data cente...	marketrent	positive	positive	positive	—
Meta Furious Over Bombshell Smart Glasses Revelation	IKeepItLayingAround	negative	negative	negative	—
Meta Deletes Face-Recognition System From Its Smart Glasses App After ...	Plastic_Ninja_9014	positive	positive	positive	—
Meta So Desperate for Compute That It's Building "Data Centers" That A...	IKeepItLayingAround	negative	negative	negative	—
Hackers tricked Meta AI into letting them take over high-profile accou...	Few_Baseball_3835	negative	negative	negative	—
Silicon Valley including Meta has embraced Maga politics, says Nick Cl...	ArgentineBeauty	neutral	neutral	neutral	—
Meta Says 20,000 Instagram Accounts Hacked via AI Tool Abuse	rkhunter_	negative	negative	negative	—
Signal, DuckDuckGo, and NordVPN threaten to exit Canada if metadata su...	rkhunter_	negative	negative	negative	—
Two thirds of our customers' fraud cases start on Meta, Lloyds says	MarvelsGrantMan136	negative	negative	negative	fraud
Meta made its own AI-generated clickbait news feed Meta said it woul...	Hrmbee	neutral	neutral	neutral	—

Title	Author	Ensemble	VADER	LR	Risk terms
Women in Brussels 'filmed without their knowledge' by men wearing Meta...	marketrent	positive	positive	positive	—
Meta Silently Added Face-Recognition Code for Its Smart Glasses to Mil...	Just-Grocery-2229	positive	positive	positive	—
Meta Silently Added Face-Recognition Code for Its Smart Glasses to Mil...	Hrmbee	positive	positive	positive	—
Meta's stock sinks on report company could raise tens of billions of d...	Logical_Welder3467	neutral	neutral	neutral	—
Meta weighs big equity raising to finance AI infrastructure, FT report...	Franco1875	neutral	neutral	neutral	—
Meta's Facial Recognition Plans for Smart Glasses Are Worse Than We Th...	Plastic_Ninja_9014	negative	negative	negative	—
Meta weighs big equity raising after blockbuster Google deal	joe4942	positive	positive	neutral	—
Meta putting up tents across the US to house AI servers, like 'a scene...	Logical_Welder3467	negative	negative	negative	—
Meta AI's recent hack is a terrifying wake-up call for anyone who puts...	Plastic_Ninja_9014	negative	negative	negative	—
Meta workers can opt out of being tracked at work - but only for half ...	ArgentineBeauty	neutral	neutral	neutral	—
The AI race has reached the Mad Max phase. Meta, Tesla putting up wate...	GeneReddit123	negative	negative	negative	—

Title	Author	Ensemble	VADER	LR	Risk terms
A Louisiana state senator helped secure Meta's largest datacenter. The...	Just-Grocery-2229	positive	positive	positive	—
How Meta, JPMorgan, and others track AI use, and why it's getting comp...	Plastic_Ninja_9014	neutral	neutral	neutral	—
Mark Zuckerberg wants Meta agents to "run your whole business"	joe4942	neutral	neutral	neutral	—
Meta scales back plan to track workers' clicks and keystrokes to train...	captainbbeard	neutral	neutral	neutral	—
Meta AI chatbot enabled hackers to access others' Instagram accounts	Repulsive-Hurry8172	neutral	neutral	neutral	—
Instagram users locked out after Meta AI abused to steal accounts	cf18	negative	negative	negative	—
Meta's AI training effort is capturing employee emails and browsing hi...	AdSpecialist6598	neutral	neutral	neutral	—
Hackers trick Meta AI support bot to infiltrate Obama White House Inst...	ArgentineBeauty	positive	positive	positive	—
Zuckerberg's superyacht lands in Seattle hours after Meta announces bi...	marketrent	neutral	neutral	neutral	—

9. NLP techniques demonstrated (rubric)

Minimum **3 basic** + **3 advanced** techniques — implementation mapping for your written report.

Basic techniques

- **Preprocessing** (tokenize, stem, lemmatize, URL removal)
- **Named entity recognition** (spaCy / NLTK)
- **Part-of-speech tagging**
- **TF-IDF + Logistic Regression classification**
- **Sentiment ensemble** (VADER + LR majority vote)
- **Embedding-based topic clustering** (KMeans)
- **Regex information extraction** (hashtags, mentions, risk lexicon)

Advanced techniques

- **Foundation LLM** (OpenAI-compatible briefing)
- **Retrieval-augmented generation** (top-k posts)
- **Prompt engineering** (structured analyst template)
- **Chain-of-thought prompting**
- **LLM-as-judge / automated evaluation**
- **Ensemble methods** (sentiment + multi-signal crisis)
- **Agentic ReAct-style orchestration**
- **Ethics & fairness disclaimers**

10. Agent orchestration & evaluation

Agent status: completed

ReAct-style trace

Step	Thought	Action	Observation
1	Need Reddit posts about the brand/topic.	collect_posts	Starting data collection.
2	Collected 45 posts.	collect_posts	Received 45 posts.
3	Raw text must be normalized for NLP.	preprocess	Running tokenizer pipeline.
4	Preprocessing complete.	preprocess	45 posts prepared.
5	Extract hashtags, mentions, risk keywords.	extract	Regex IE.
6	Run NER and POS for entity-aware monitoring.	ner_pos	spaCy/NLTK.
7	Classify sentiment with VADER + LR ensemble.	sentiment	Ensemble vote.
8	Cluster topics with embeddings.	topics	KMeans clustering.
9	Compute temporal trends.	trends	Daily aggregation.
10	Score crisis risk from sentiment + engagement + trends.	crisis	Crisis formula.
11	Retrieve relevant posts for grounded LLM briefing (RAG).	rag	Query=meta
12	Generate executive briefing with CoT LLM.	briefing	LLM or fallback.

Automated evaluation

Sentiment agreement (VADER vs LR): 97.8% — VADER and LR agree on 44/45 posts (98%). Use manually labelled gold set for formal accuracy in your report.

Briefing judge score: 70/100 (error_fallback)

LLM judge unavailable; use manual rubric in report.

11. Ethics, limitations & references

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- Public Reddit text only; no private user data.
- Crisis scores are decision-support estimates, not legal or PR advice.
- Sentiment models can misclassify sarcasm and domain-specific language.
- For quantitative evaluation on labelled data, extend `backend/evaluation.py`.
- Full NLP technique reference: `docs/NLP_TECHNIQUES.md`

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